

Inverse of 2x2 matrices

Basic skills

Q1. a) $\begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc$
 $\begin{vmatrix} 3 & 1 \\ 5 & -2 \end{vmatrix} = 3 \times -2 - 1 \times 5 = -6 - 5 = -11$

b) $\begin{vmatrix} 1 & 3 \\ 0 & 5 \end{vmatrix} = 5 - 0 = 5$

c) $\begin{vmatrix} 5 & 1 \\ 10 & 2 \end{vmatrix} = 10 - 10 = 0$

Q2. a) $\begin{pmatrix} 5 & 3 \\ 1 & 2 \end{pmatrix} \xrightarrow[\text{swap positions}]{\text{swap signs}} \begin{pmatrix} 2 & -3 \\ -1 & 5 \end{pmatrix}$

b) $\begin{pmatrix} 2 & 2 \\ -1 & 1 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & -2 \\ 1 & 2 \end{pmatrix}$

c) $\begin{pmatrix} 4 & -1 \\ 0 & 2 \end{pmatrix} \rightarrow \begin{pmatrix} 2 & 1 \\ 0 & 4 \end{pmatrix}$

Q3. a) $\begin{pmatrix} 3 & 0 \\ 2 & 1 \end{pmatrix}$ $\begin{vmatrix} 3 & 0 \\ 2 & 1 \end{vmatrix} = 3 - 0 = 3$
 $\text{Adj} = \begin{pmatrix} 1 & 0 \\ -2 & 3 \end{pmatrix}$

So inverse is $\frac{1}{3} \begin{pmatrix} 1 & 0 \\ -2 & 3 \end{pmatrix} \leftarrow \frac{1}{|M|} \text{adj}(M)$

b) $\begin{pmatrix} 5 & 1 \\ 2 & 1 \end{pmatrix}$ $\left| \begin{pmatrix} 5 & 1 \\ 2 & 1 \end{pmatrix} \right| = 5 - 2 = 3$

$$\text{Adj} = \begin{pmatrix} 1 & -1 \\ -2 & 5 \end{pmatrix}$$

so inverse is $\frac{1}{3} \begin{pmatrix} 1 & -1 \\ -2 & 5 \end{pmatrix}$

c) $\begin{pmatrix} 5 & 4 \\ 2 & -3 \end{pmatrix}$ $\left| \begin{pmatrix} 5 & 4 \\ 2 & -3 \end{pmatrix} \right| = -15 - 8 = -23$

$$\text{Adj} = \begin{pmatrix} -3 & -4 \\ -2 & 5 \end{pmatrix}$$

So Inverse is $\frac{1}{-23} \begin{pmatrix} -3 & -4 \\ -2 & 5 \end{pmatrix} = \frac{1}{23} \begin{pmatrix} 3 & 4 \\ 2 & -5 \end{pmatrix}$

Q4. when $|M| = 0$ as dividing by it
so zero determinant matrices are not invertible.

Competency skills

Q5. Let $M = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ $M^{-1} = \begin{pmatrix} e & f \\ g & h \end{pmatrix}$

then $MM^{-1} = I \Rightarrow \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} e & f \\ g & h \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$

$$\begin{aligned} \text{So } ae + bg &= 1 & af + bh &= 0 \\ ce + dg &= 0 & cf + dh &= 1 \end{aligned}$$

↑ want e, g, f, h
(in terms of a, b, c, d)

$$\text{So } ce + \frac{bc}{a}g = \frac{c}{a}$$

$$- \quad ce + dg = 0$$

$$\left(\frac{bc}{a} - d\right)g = \frac{c}{a}$$

$$g = \frac{c}{bc - ad} = \frac{-c}{ad - bc}$$

$$e = -\frac{d}{c}g = \frac{d}{ad - bc}$$

Similarly

$$cf + dh = 1$$

$$- \quad cf + \frac{bc}{a}h = 0$$

$$\Rightarrow \left(d - \frac{bc}{a}\right)h = 1 \Rightarrow h = \frac{a}{ad - bc}$$

$$f = -\frac{b}{a}h = \frac{-b}{ad - bc}$$

$$\text{So } M^{-1} = \begin{pmatrix} e & f \\ g & h \end{pmatrix} = \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$$

as required

Q6. $M = \begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix}$

a) $|M| = 6 - 1 = 5$ $\text{Adj}(M) = \begin{pmatrix} 3 & -1 \\ -1 & 2 \end{pmatrix}$

so $M^{-1} = \frac{1}{5} \begin{pmatrix} 3 & -1 \\ -1 & 2 \end{pmatrix}$

b) $M \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 3 \\ 4 \end{pmatrix} \Rightarrow \begin{pmatrix} x \\ y \end{pmatrix} = M^{-1} \begin{pmatrix} 3 \\ 4 \end{pmatrix}$

so $\begin{pmatrix} x \\ y \end{pmatrix} = \frac{1}{5} \begin{pmatrix} 3 & -1 \\ -1 & 2 \end{pmatrix} \begin{pmatrix} 3 \\ 4 \end{pmatrix} = \frac{1}{5} \begin{pmatrix} 5 \\ 5 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$

Q7. Let $C = (AB)^{-1}$

Then $C(AB) = I$

so $C = B^{-1}A^{-1}$ as $B^{-1}A^{-1}AB = B^{-1}B = I$

so $(AB)^{-1} = B^{-1}A^{-1}$

Q8. $\begin{pmatrix} k & 2k \\ 3 & k \end{pmatrix}$ has no inverse so determinant zero

$\Rightarrow k \times k - 3 \times 2k = 0 \Rightarrow k^2 - 6k = 0$

so $k(k-6) = 0 \Rightarrow k = 0 \text{ or } 6$

Q9. $A = \begin{pmatrix} 3 & 5 \\ 1 & 3 \end{pmatrix} \quad AB = \begin{pmatrix} 5 & 3 \\ 0 & 2 \end{pmatrix}$

Algebraically $\cancel{A}^{-1}AB = \cancel{A}^{-1} \begin{pmatrix} 5 & 3 \\ 0 & 2 \end{pmatrix}$

need $A^{-1} = \frac{1}{|A|} \text{adj}(A) = \frac{1}{9-5} \begin{pmatrix} 3 & -5 \\ -1 & 3 \end{pmatrix} = \frac{1}{4} \begin{pmatrix} 3 & -5 \\ -1 & 3 \end{pmatrix}$

so $B = \frac{1}{4} \begin{pmatrix} 3 & -5 \\ -1 & 3 \end{pmatrix} \begin{pmatrix} 5 & 3 \\ 0 & 2 \end{pmatrix} = \frac{1}{4} \begin{pmatrix} 15 & -1 \\ -5 & 3 \end{pmatrix}$

Advanced skills

Q10. Let $M = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$

Then $\begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \end{pmatrix} = \begin{pmatrix} 0 & 5 & 2 & 8 \\ 0 & 4 & 3 & 10 \end{pmatrix}$

$\Rightarrow a=5, c=4, b=2, d=3$

but also $a+b=8, c+d=10$

no feasible solution so cannot exist

Q11. If $M = M^{-1}$

$\Rightarrow \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \frac{1}{ad-bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$

$$\Rightarrow a = \frac{d}{ad-bc}, \quad b = \frac{-b}{ad-bc}$$

$$c = \frac{-c}{ad-bc}, \quad d = \frac{a}{ad-bc}$$

$$\Rightarrow (ad-bc)a - d = 0$$

$$(ad-bc+1)b = 0$$

$$(ad-bc+1)c = 0$$

$$(ad-bc)d - a = 0$$

so either: • $b=0, c=0$ and $a^2d-d=0$
 $d^2a-a=0$

$$\text{so } ad=1$$

• $ad-bc=-1$ then $-a-d=0$

$$\text{so } d=-a$$

so either

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}, \begin{pmatrix} a & b \\ c & -a \end{pmatrix}$$

↑
rotation
180°

for a, b, c

$$\text{s.t. } a^2 - bc = -1$$

Q12. If uninvertible $|M|=0$ so $ad-bc=0$

then $M^2 = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} a^2+bc & ab+bd \\ ac+dc & cb+d^2 \end{pmatrix}$

As $ad=bc$ $M^2 = \begin{pmatrix} a^2+ad & b(a+d) \\ (a+d)c & ad+d^2 \end{pmatrix}$
 $= (a+d) \begin{pmatrix} a & b \\ c & d \end{pmatrix}$

so by induction $M^n = M M^{n-1}$
 $= M (a+d)^{n-2} M$
 $= (a+d)^{n-2} M^2$
 $= (a+d)^{n-1} M$

Q13. He needs to say when $|A| \neq 0$

Q14. $ABCD = I$

$\Rightarrow \cancel{A}^{-1}ABCD = \cancel{A}^{-1}I \Rightarrow BCD = \cancel{A}^{-1}$

$\Rightarrow BCD\cancel{D}^{-1}\bar{C}' = \cancel{A}^{-1}\bar{D}'\bar{C}' \Rightarrow B\cancel{C}' = \bar{A}\bar{D}'\bar{C}'$

$\Rightarrow B = \bar{A}'\bar{D}'\bar{C}' = (DA)^{-1}\bar{C}' = (CDA)^{-1}$

so $B^{-1} = CDA$